

Production *AI* systems, shipped.

Gen AI Developer · Backend Engineer

A showcase of **RAG / GraphRAG pipelines**, **LLM agents**, fine-tuned models, and **FastAPI** backends I've designed, deployed, and operated — with real users, real metrics, and full ownership of deployment, CI/CD, and monitoring.

5+

YEARS SHIPPING

50K+

MONTHLY QUERIES

1st

META AI HACKATHON

+28%

ACC. VS FLAT RAG

LERN

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00 About & services

freelance · contract · full-time

I'm a Gen AI developer and backend engineer with **5+ years** of software experience, focused on getting LLM systems into production — not demos, but systems that serve real users and keep working under load. I architected a GraphRAG chatbot serving **50K+ monthly queries**, won **1st place at the Meta AI / AMD / PyTorch Hackathon**, co-authored peer-reviewed research (LERN 2026), and contributed to Meta's open-source PyTorch tooling. I hold an **M.S. in Computer Engineering from Arizona State University**. I take AI ideas from a blank repo to a deployed, observable, maintainable product.

RAG & GraphRAG systems

Chatbots and search that cite your data — chunking, embeddings, vector + graph retrieval, hallucination guardrails. LangChain · Neo4j · Pinecone · pgvector.

LLM agents & automation

Agentic workflows that use tools and finish multi-step tasks. LangGraph orchestration, MCP & tool use, structured output, on Claude & OpenAI.

LLM fine-tuning

Custom domain models — LoRA/QLoRA & GRPO on Llama 3, Qwen 3, Phi-4 — with synthetic-data pipelines and proper evaluation.

Python / FastAPI backends

Async & streaming endpoints, Pydantic validation, auth, caching, Prometheus metrics — Dockerized and ready to deploy.

Cloud deployment & MLOps

AWS / Azure / GCP, CI/CD, containerization, model versioning, A/B testing, and monitoring you can trust.

Computer vision

Detection & segmentation on drone / ground / street-view imagery — model benchmarking and reproducible training.

01 Selected work

case study 1 of 5

PRODUCTION GRAPHRAG · MULTILINGUAL · 50K+ QUERIES/MO

Arizona WaterBot

A statewide multilingual chatbot answering water-resource questions with relationship-aware, grounded retrieval.

50K+ monthly queries

+28% accuracy

AWS · Azure

ASU engagement



GraphRAG retrieval pipeline — relationship-aware context beats flat vector RAG on accuracy.

THE CHALLENGE

Public users needed accurate, multilingual answers about water policy — where flat RAG hallucinated and missed relationships between entities.

WHAT I BUILT

A GraphRAG pipeline (LangChain + Neo4j) with hybrid vector + graph traversal, an async FastAPI backend with streaming, and output guardrails. Spanish support via a multilingual NLP layer.

IMPACT

Served **50K+ monthly queries** on AWS & Azure with **+28% accuracy** over flat-RAG baselines; basis of a peer-reviewed LERN 2026 paper.

Stack: LangChain · Neo4j · OpenAI · FastAPI · PostgreSQL · AWS · Azure · Docker

Startup Intelligence Engine

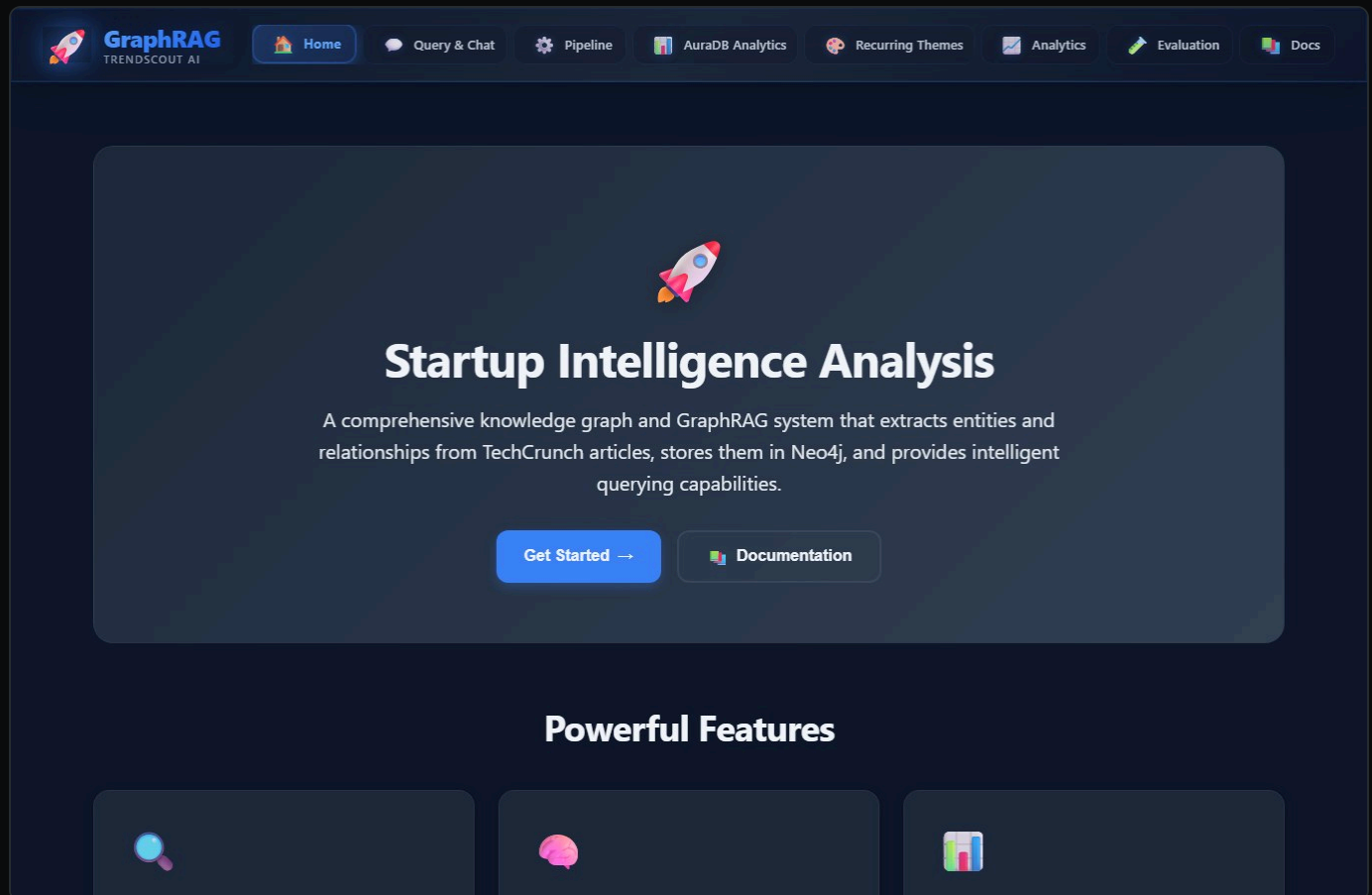
Extracts startup intelligence from TechCrunch into a Neo4j knowledge graph, with natural-language querying over companies, investors & funding.

200× faster cached

70%+ test coverage

40+ endpoints

live demo



Live application — startup-intelligence-analysis-app.vercel.app

THE CHALLENGE

Turn unstructured news into queryable intelligence — entities, relationships, and funding trends — fast enough for interactive use.

WHAT I BUILT

A 5-phase pipeline: scrape → GPT-4o entity extraction → enrichment → Neo4j graph → embeddings + community detection. FastAPI backend with JWT auth, rate limiting, Redis cache, and 15+ Prometheus metrics.

IMPACT

200× faster cached responses (2000ms→10ms), **70%+ test coverage** across 150+ tests, GitHub Actions CI/CD.

Stack: FastAPI • Neo4j • GPT-4o • Redis • React • Docker
[Intelligence-Analysis-App](#) ↗

[github.com/shankerram3/Startup-](https://github.com/shankerram3/Startup-Intelligence-Analysis-App)

🏆 1ST PLACE • SYNTHETIC DATA • LORA FINE-TUNING

Meta AI Hackathon — Reasoning Models

Won 1st place at the Meta AI / AMD / PyTorch Synthetic Data Challenge: a synthetic-data pipeline + fine-tuned models for relational reasoning.

1st place

16K+ LOC

95%+ coverage

4+ models



Synthetic-data → reward-shaping → fine-tune loop; mitigated an alphabetical-sorting reasoning bias.

THE CHALLENGE

Generate high-quality training data for complex relational-reasoning tasks (character relationships, seating arrangements) and lift model accuracy.

WHAT I BUILT

A multi-stage synthetic-data pipeline with structured-JSON enforcement, GRPO reward functions, and LoRA/SFT fine-tuning across Llama 3 70B, Qwen3-Next 80B MoE, and Phi-4.

IMPACT

1st place overall; identified & mitigated a major LLM reasoning flaw, improving logical consistency. 16K+ LOC, 95%+ coverage.

Stack: PyTorch • LoRA • GRPO • Llama 3 • Qwen 3 • Phi-4 • Unsloth

[technical deep dive ↗](#)

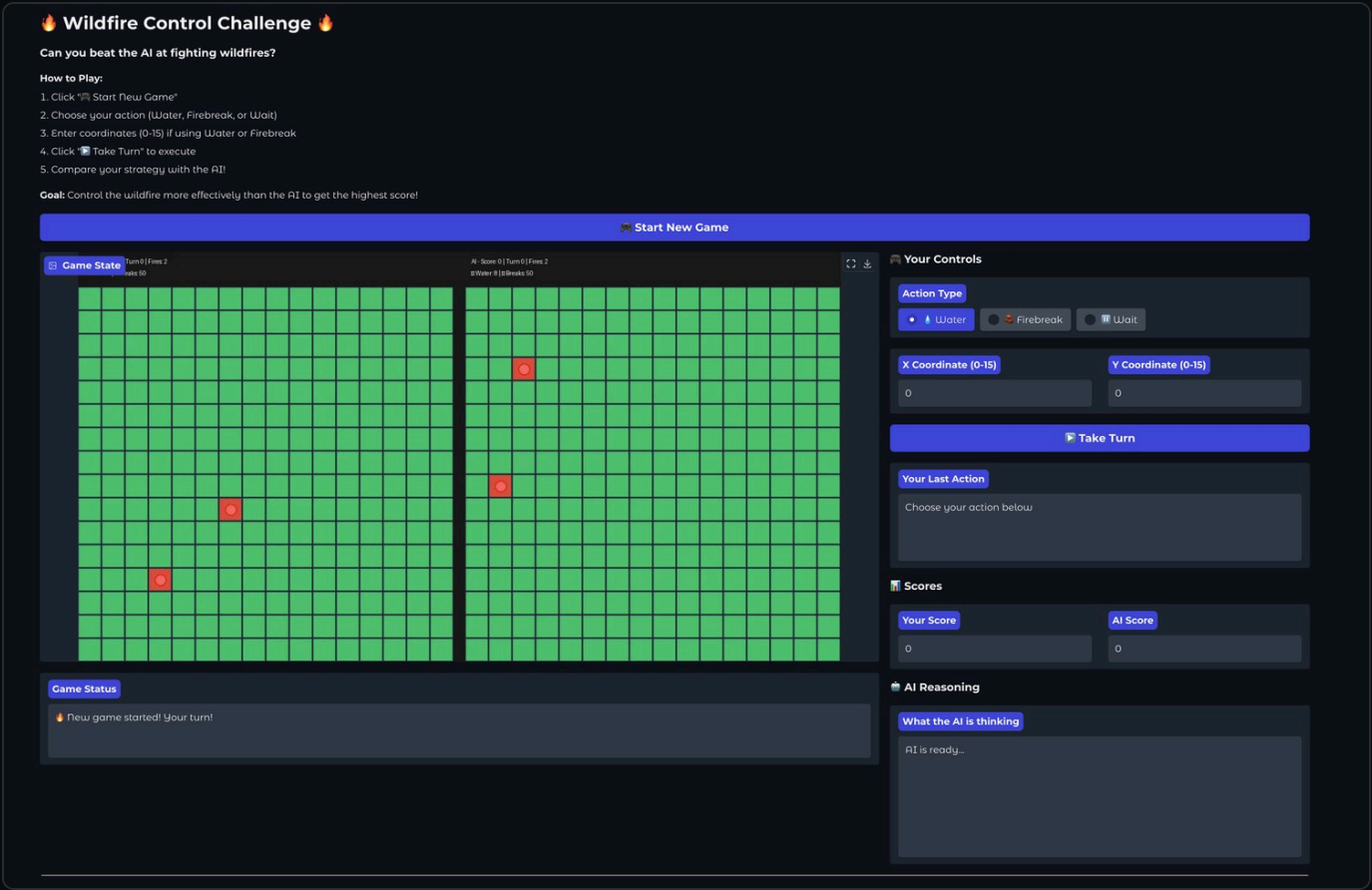
Wildfire Control RL

A physics-informed wildfire-containment environment + RL agent — contributed to Meta's open-source PyTorch OpenEnv, with an interactive human-vs-AI game.

Final Four

16K+ demos

OpenEnv PR



Interactive wildfire-containment game — human vs. trained RL agent.

WHAT I BUILT

An RL environment with fire-spread physics; trained Llama 3.2 with LoRA on 16,000+ demonstrations using GRPO + behavior cloning on AMD MI100.

OPEN SOURCE

Contributed the environment to Meta PyTorch OpenEnv with documentation and unit tests; deployed inference to Hugging Face Spaces.

STACK

PyTorch · Llama 3.2 · LoRA · GRPO · Hugging Face

MakerBOT — AI Tutor

Claude returns explanations *plus* executable React components, safely rendered in a sandboxed iframe — real-time interactive STEM visualizations.



02 More & how to reach me

ALSO BUILT

NomadSync

Collaborative AI trip planner — a LangGraph agent with trip-memory extraction, plan versioning, live flight data, and real-time multiplayer over SSE.

LangGraph · OpenAI · Express · MongoDB · React

RagPiper

A flexible, production-ready GraphRAG engine with a pluggable architecture for swapping retrievers, stores, and models.

Python · GraphRAG · Neo4j

Graph Processing at Scale

Large-scale graph pipeline (NYC taxi data) with PageRank & BFS on Neo4j GDS, orchestrated on Kubernetes + Kafka.

Neo4j · Kubernetes · Kafka · AWS · Terraform

AI-Image Classification

ResNet50v2 / EfficientNetV2 model distinguishing AI-generated vs. real images, evaluated with AUC/F1 across CFAKE & ArtiFact.

PyTorch · CNNs · Transfer Learning

TECHNICAL SKILLS

LLMs & Agents — Claude API, OpenAI, Gemini, LangGraph, MCP, LoRA/QLoRA, GRPO

RAG & Retrieval — GraphRAG, LangChain, LlamaIndex, Neo4j, Pinecone, ChromaDB, pgvector

Backend & Cloud — Python, FastAPI, Node.js, Docker, Kubernetes, AWS, Azure, GCP, CI/CD

Data & MLOps — PostgreSQL, MongoDB, Redis, W&B, MLflow, Prometheus

ML & CV — PyTorch, Hugging Face, YOLOv8-seg, SAM 2, HAWP

EXPERIENCE

ML / Computer Vision Engineer

Mesa Associates, Inc.

Knoxville, TN · Jan 2026 – Present

AI Engineer / Full-Stack Developer

Arizona Water Innovation Initiative · ASU

Tempe, AZ · May 2024 – Dec 2025

Software Engineer — AI Integration

PIT Solutions Pvt. Ltd.

Trivandrum, India · 2022 – 2023

Software Engineer & Co-Founder

MakerLabs Edutech

Kochi, India · 2020 – 2022

Have an AI idea? Let's *ship* it.

Available for freelance & contract work — RAG, agents, fine-tuning, and FastAPI backends. Also open to full-time ML & software engineering roles.

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